

1. Workload

2. ISA & ABI

3. Compute Org

4. Memory & NoC

5. Power Intent

6. Compiler/RT

7. Physical Node

8. Verification



CHAPTER STACK SCOPE — ACTIVE LAYERS 3-7 (SUBSYSTEM TO LAYOUT)

Environments, Tool Interfaces & Sandbox Isolation

1. Agent Sandboxes

- Containerized Tool State
- Ephemeral Execution Pods
- Resource Quota Enforcer
- Security Isolation Boundary

2. Tool Adapters

- Structured JSON/CLI Wrapper
- gem5 Simulator Adapter
- Verilator Harness Interface
- OpenROAD Script Driver

3. Execution Harness

- Parallel Job Scheduler
- Log Parser & Extraction
- Deterministic Seed Control
- Error Timeout Intercept

4. Safe Handoff

- Execution Provenance Card
- Reproducibility Check
- Tool Environment Freeze
- Handoff Signoff

Environment Recovery & Error Intercept Feedback